



Fig. 5: Assembled unit placed on the handle



Fig. 6: Hall sensor and magnet fixed on the front wheel of the motorbike

wheeler over-speeds.

The circuit is powered by the 12V battery of the two-wheeler. IC 7805 (IC7) is used to provide regulated 5V

DC to IC1 and IC2. Capacitors C8 and C9 bypass the ripples present in the supply. LED2 acts as the power indicator and R12 limits the current through LED2.

Construction

An actual-size, single-side PCB for the over-speed alarm is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error.

Now calibrate the circuit. Let us say you want to set the speed limit for the two-wheeler to 40 kmph. For this, first find out the corresponding rotational frequency (revolutions per second) of the wheel using the formula:

$$\frac{\text{Speed (in kmph)} \times 10^3}{\text{Diameter (in cm)} \times \pi \times 3600} = \text{Hz}$$

For a 56cm dia. front wheel and a speed of 40 kmph, it is calculated to be 6 Hz. Apply a 6Hz squarewave

signal from a function generator to pin 2 of pulse generator IC2 via resistor R1. Adjust VR1 such that LED1 glows brightly. Now adjust presets VR2, VR3 and VR4 to sound the speaker. Now if you reduce the input signal frequency at IC2, LED1 should go off and the audio alarm stop.

House the entire circuit in a suitable cabinet with the Hall sensor connected using a three-wire cable. Fix the unit on the handle of the two-wheeler (refer Fig. 5) and the Hall sensor (A1302) on the shock absorber of the front wheel. Also fix a magnet on the drum of the front wheel (refer Fig. 6) so that a pulse is generated by IC2 whenever the magnet comes in line with the Hall sensor as the wheel rotates.

The circuit is now ready for use. It will sound an alarm whenever you cross the speed limit of 40 kmph. The alarm will stop as soon as you slow down the two-wheeler below the set limit. ●

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